

AMKean-Vasovmodel

$$\bullet \quad dX_t^i = \sum_{j=1}^N a(X_t^i - X_t^j) dt + \sigma dW_t^i$$

- What is the behaviour depending on a , σ , N ?

• For $a < 0$:

◦ $Y_t^i = X_t^i - \frac{1}{N} \sum_i X_t^i$ is an Ornstein-Uhlenbeck process

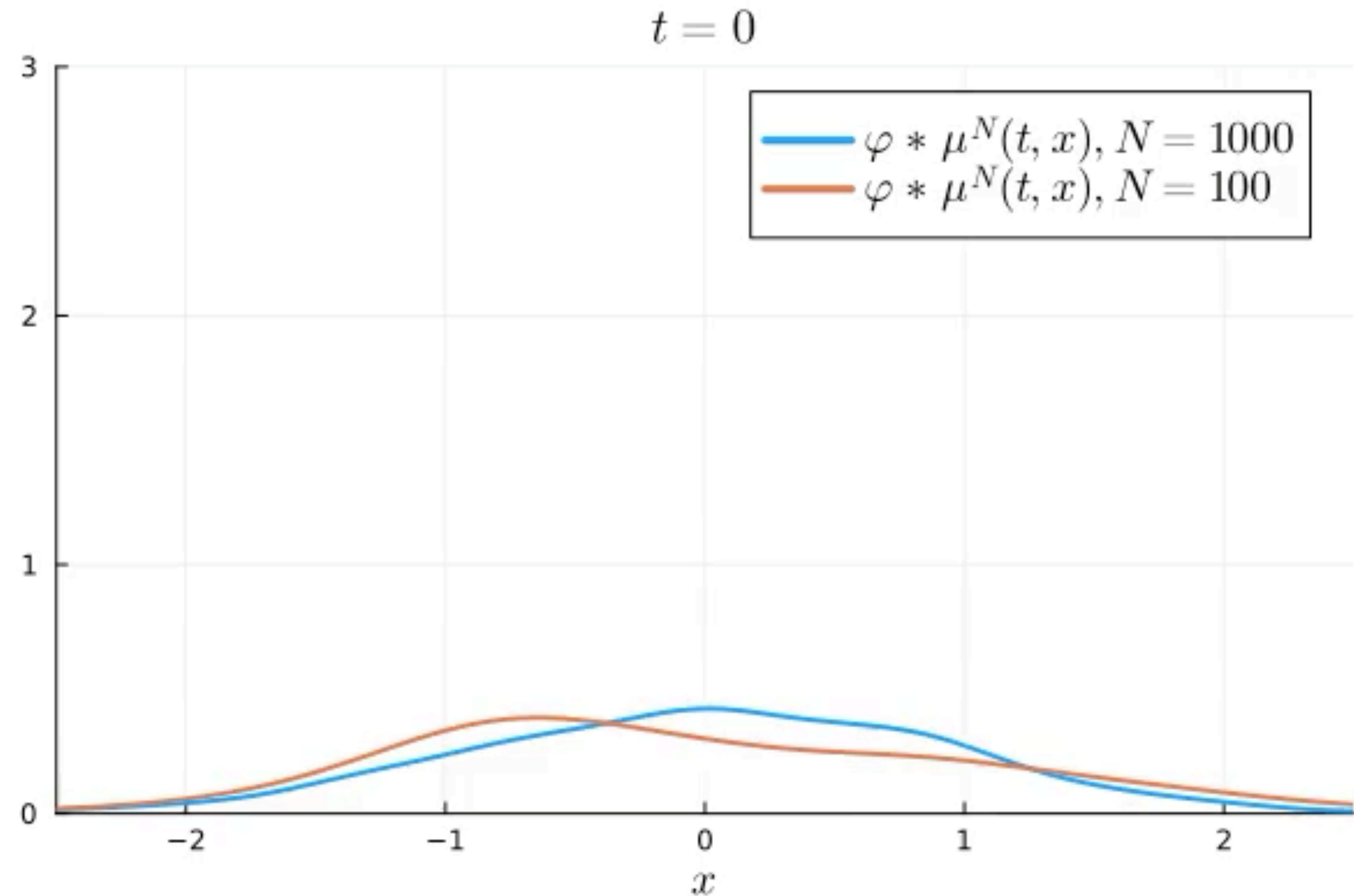
$$\circ \quad \frac{1}{N} \sum_i X_t^i = \frac{1}{\sqrt{N}} \hat{W}_t$$

A McKean-Vlasov model

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A McKean-Vlasov model

- $dX_t^i = \sum_{j=1}^N a(X_t^i - X_t^j)dt + \sigma dW_t^i$
- Empirical measure $\mu_t^N = \frac{1}{N} \sum_i \delta_{X_t^i}$



$$a = -1/N, \sigma = 0.1$$